Saturation as a Result: Ensembling, Length-Prior Decoding, and the Data Ceiling in Arabic Sentence Segmentation

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Abstract

We describe our submission to the AraSeg 2026 Arabic sentence-segmentation shared task, entered in all four task variants and both the closed and open tracks. We frame segmentation as binary token classification, fine-tune Arabic encoders, combine them by probability averaging, and decode with a semi-Markov dynamic program that adds a train-fit segment-length prior and forces structurally certain boundaries. The system ranked first on all four closed-track tasks during the development phase. Our central contribution is a controlled study of where these gains *stop*: across roughly twenty additions—larger and genre-specialised encoders, more seeds, corruption augmentation, adversarial and label-smoothing training, and external-data pretraining—held-out performance saturates. We trace this to data rather than capacity, and quantify it: a training-set *scaling curve* (three seeds) that is flat on the easiest task (α=1.37, at its asymptote) but still rising on the hardest (α=0.66, $\sim$1.9 F_1 below an 86.4 asymptote, CI confirmed); a model-free check showing gold labels are 99%-consistent for repeated contexts, so the model's 0.85–3.10% perceived ambiguity is mostly *sparsity*, not noise; a *coverage* measurement showing only 5% of no-punctuation boundary contexts are seen in training; and a Krogh–Vedelsby decomposition pinning the ensemble's small head-room on a voter *diversity* of just 14%. We pair this with a bootstrap procedure for selecting configurations under a small development set, and an open-track result in which external pretraining helps only through ensemble diversity, and only on the hardest condition.
Code, experiment log, and reproduction pipeline: `https://github.com/omarsaqr12/arabic-sentence-segmentation`

1 Introduction

The AraSeg 2026 shared task asks systems to predict, for every whitespace token of an Arabic document, whether a sentence boundary follows it. Four variants cross the availability of paragraph structure with the availability of punctuation: PA (paragraphs and punctuation), NoPnx-PA (paragraphs, no punctuation), NP (punctuation, no paragraphs), and NoPnx-NP (neither). The *closed* track restricts fine-tuning data to the 174-document AraSeg training set; the *open* track permits any public data. Performance is per-document binary precision, recall, and F_1 on boundary labels, macro-averaged over documents, so short documents and rare genres carry equal weight.

A natural strategy—fine-tune a strong encoder, ensemble, and decode—reaches a high score quickly. The interesting scientific question is what happens next: why do twenty subsequent, individually reasonable improvements fail to move the held-out number—and what, eventually, does? We answer this directly and treat the answer (data sparsity caps every conventional axis; only *bidirectional* architecture diversity buys measurable room) as the paper's main result.

Contributions. (1) A strong, reproducible closed-track system (encoder ensemble plus semi-Markov length-prior decoding) that led all four development-phase leaderboards (§3–5). (2) A controlled *saturation study* (§6) with held-out results for $\sim$20 attempts. (3) Four convergent diagnostics—scaling, coverage, voter correlation, and error typology—that locate the ceiling in *data*, not architecture, which we make quantitative with a power-law scaling fit and a calibration-grounded irreducible-error floor (§7–7.1). (4) A bootstrap config-selection procedure for small development sets, which we validate on held-out data and which changes two of our four submissions (§8). (5) An open-track study showing external boundary-recovery pretraining helps only via ensemble diversity, and only on NoPnx-NP (§10).

2 Task and Data

The corpus contains 174 training, 222 development, and 262 test documents, averaging $\sim$700 tokens, drawn from eight genres. Tokens are whitespace separated; punctuation marks are their own tokens; paragraph breaks are literal newline tokens in PA variants; and the gold label sits *on* the sentence-final token. The four variants are not independent: the no-punctuation inputs are exact deletion subsequences of the punctuated inputs for the same document identifier.

Three properties shape modelling. The comma is only sometimes a boundary (e.g. in *hadīth* transmission chains), so punctuation rules cannot saturate the punctuated tasks. The token before a paragraph newline is a boundary with 100% precision on development data. And the final non-newline token of every document is a boundary in 396/396 train+dev documents, a safe forced prediction. We additionally report two issues to the organisers rather than exploit them: the released `eval.py` prints precision and recall under swapped labels (F_1 correct), and aligned cross-variant document identifiers create a punctuation-projection leakage vector we flag for blind-test de-aliasing.

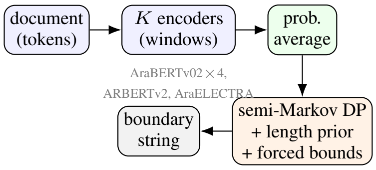

Figure 1: System pipeline: K fine-tuned encoders (K=3–6 per task after selection, §8, 10), per-token probability averaging, then semi-Markov dynamic-programming decoding with a train-fit length prior and forced (pre-newline / document-final) boundaries.

Encoder	PA	N-PA	NP	N-NP
AraBERTv02	**94.81**	**86.43**	**92.92**	**83.07**
AraELECTRA	94.43	85.73	92.18	82.82
ARBERTv2	94.34	84.94	91.95	82.11
larger/other (proxy F_1): < base on $4/4$				
AraBERT-large, XLM-R, mmBERT, CAMeLBERT-MSA/CA				

Table 1: Encoder comparison (dev macro-F_1). N-PA=NoPnx-PA, N-NP=NoPnx-NP.

3 System

Figure 1 summarises the pipeline.

3.1 Base model

We treat segmentation as binary token classification on a fine-tuned encoder. Each whitespace token is labelled on its *last* sub-word (others masked); paragraph newlines map to a `[PAR]` token excluded from the loss and forced to non-boundary at inference. We use class-weighted cross-entropy (boundary base rate 0.08–0.11), overlapping word windows of length 180 (stride 90), and average window probabilities at inference.

3.2 Encoder choice

Table 1 compares three Arabic encoders. AraBERTv02 [1] wins all four tasks over AraELECTRA [2] and ARBERTv2 [3]. Critically, *larger* and differently pretrained encoders—AraBERT-large, XLM-R-large [5], mmBERT, and the CAMeLBERT family [4]—all underperform the base model. This is the first hint of a data-limited regime: with 174 documents, additional capacity cannot be specified.

3.3 Ensemble and decoding

We average per-token boundary probabilities over six voters (four AraBERTv02 seeds, ARBERTv2, AraELECTRA); the per-task pools submitted are later refined by the selection procedures of §8 and §10 to 3–6 voters. Rather than thresholding independently, we choose the boundary set B maximising

$$\sum_{i\in B}\log p_i + \sum_{i\notin B}\log(1-p_i) + \lambda \sum_{s\in \mathrm{seg}(B)} \log P_{\mathrm{len}}(|s|), \quad (1)$$

Component (cumulative)	PA	NoPnx-NP
Single AraBERTv02	94.81	83.07
+ 6-encoder ensemble	94.94	84.16
+ length-prior DP, forced bounds	95.15	84.37
+ document-adaptive prior	95.15	84.53
decode leave-one-out (NoPnx-NP, from 84.53*):*		
− document-adaptive		$84.37\ (-0.16)$
− recall bias		$84.05\ (-0.48)$
− length prior		$83.41\ (-1.12)$

Table 2: Top: cumulative ablation (dev macro-F_1); ensembling dominates on the hard task. Bottom: decode leave-one-out—the length prior is the largest decode component over the *calibrated* ensemble (cf. its near-zero effect over single models, §6).

where P_{len} is a segment-length distribution fit on *training* (closed-legal), solved by an $O(nL_{\max})$ dynamic program with forced pre-newline and document-final boundaries. A two-pass *document-adaptive* variant re-estimates each document's length distribution from a first-pass decode. Table 2 ablates each component.

4 Experimental Setup

Closed-track models are fine-tuned on one RTX 4080 (16 GB) with `torch` 2.5.1 / `transformers` 4.57; a `torch` 2.12 environment loads checkpoints shipped without `safetensors`. All hyperparameters—encoders, thresholds, decode parameters—are selected on development; the public open-test split is held out for verification only; test inputs are used solely for prediction. Significance is by paired bootstrap over documents (5000 resamples).

5 Results

The full pipeline reaches development F_1 of 95.15/87.60/93.37/84.53 and open-test F_1 of 94.42/87.27/92.69/84.97 (closed track; the NoPnx-PA system includes the mDeBERTa voter adopted in §10); the development-to-test gap is ≤ 0.7 everywhere. During the development phase this ranked first on all four closed-track leaderboards. Table 3 contextualises this against external systems: a rule baseline, the off-the-shelf multilingual SOTA SaT-12L [6] (zero-shot $\sim$69 on every task—punctuation-sensitive but paragraph-insensitive, hence its tied PA/NP and NoPnx-PA/NoPnx-NP scores), and the organiser baseline. Against the single encoder, the ensemble-plus-decoding system is significant on three of four tasks (Table 4); on NP the $+0.24$ gain is *not* significant (p=0.14)—an early signal of saturation that our blind-test selection (§8) acts on.

System	split	PA	N-PA	NP	N-NP
Rule baseline	dev	72.2	45.2	60.2	13.4
SaT-12L (0-shot)	dev	68.9	68.5	68.9	68.5
Ours	dev	**95.2**	**87.6**	**93.4**	**84.5 / 85.1**†
Organiser baseline	test	92.8	82.8	89.7	77.8
Ours	test	**94.4**	**87.3**	**92.7**	**85.0 / 85.2**†

Table 3: External comparison (macro-F_1). Off-the-shelf SaT is far below an in-domain system; we exceed the organiser baseline by 1.3–7.4 across tasks. †closed / open track (open adds the SaT-ft voter, §10).

Attempt	Held-out test
+win-240 / CAMeLBERT / mmBERT / SaT / aug / soup voters	saturated
large / XLM-R / mmBERT / CAMeL-BERT (solo)	$<$ base 4/4
smooth+FGM, full pool	dev $+0.2$, test -0.06
greedy weighted ensemble	dev-overfit

Table 5: Representative attempts that did not beat the six-model ensemble on test.

Task	ΔF_1 vs single	p	95% CI
PA	$+0.34$	0.008	$[+0.09, +0.60]$
NoPnx-PA	$+1.17$	< 0.001	$[+0.72, +1.64]$
NP	$+0.24$	0.137	$[-0.08, +0.53]$
NoPnx-NP	$+1.46$	< 0.001	$[+1.09, +1.85]$

Table 4: Paired-bootstrap significance, full system vs. single AraBERTv02 (dev). The ensemble's gain is not significant on NP.

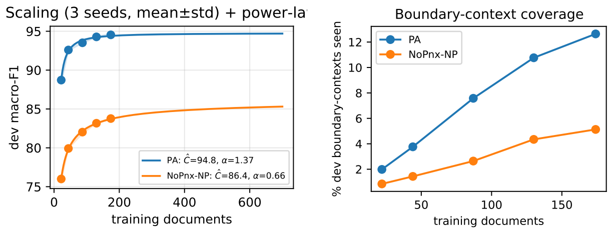

Figure 2: Left: dev F_1 vs. training-set size (3 seeds, mean±std) with power-law fits $\hat{C} - a\,n^{-\alpha}$ extrapolated to 700 docs. PA is at its asymptote (α=1.37); NoPnx-NP scales slowly toward a lower, still-unreached asymptote (α=0.66). Right: fraction of dev boundary *contexts* seen in training—only 5% for NoPnx-NP. Both locate the ceiling in data quantity.

6 What Does Not Help: A Saturation Study

Beyond the six-model ensemble we attempted $\sim$20 further improvements; none improved the held-out test result (Table 5). **Adding voters**—a window-240 model, CAMeLBERT-MSA/CA (a classical-Arabic specialist for the *hadīth* genre), mmBERT with whole-document context, zero-shot SaT [6], corruption-augmented models, and weight-space soups—left the ensemble unchanged. **Larger encoders** underperformed the base alone. **Improving every voter** with boundary label smoothing and FGM adversarial training [8] lifts a *single* model by $+0.68$ on NoPnx-NP, but full-pool retraining yields $+0.2$ dev and -0.06 test: the ensemble already captures it—so this is the recipe for a single deployable model, not the ensemble. **Tuning the combiner** by greedy per-voter weights overfit development (it zeroed two voters) and was not adopted. A consistent theme: priors help only when the probabilities they multiply are trustworthy—the length prior is a near-no-op over a weak ensemble and earns weight (λ up to 0.4) only once six *calibrated* voters back it (§7).

7 Why It Saturates: Four Diagnostics

We give four convergent measurements locating the ceiling in data, not capacity.

(i) Training-set scaling (Fig. 2). Re-training the single encoder (three seeds; std ≤ 0.4 F_1) on $\{22, 44, 87, 130, 174\}$ documents shows qualitatively different curves: PA flattens (final step $+0.26$ F_1 per 44 docs) while NoPnx-NP is still climbing ($+0.61$ per 44 docs)—a rise far exceeding the seed std, so it is signal, not noise. The hardest task is thus *data-limited*—more labelled data would still help—whereas PA is near a non-data ceiling. This both explains why larger models fail (the bottleneck is not capacity) and predicts that the lever for NoPnx-NP is effective data, which the open track supplies.

(ii) Context coverage (Fig. 2). Only 5.1% of NoPnx-NP development boundary contexts (previous,current token bigrams) appear in the 174 training documents, versus 12.6% for PA, and coverage grows slowly with training size. Most no-punctuation decisions are therefore made on contexts unseen in training—a direct, model-independent measurement of the sparsity the scaling curve reflects.

(iii) Voter redundancy (Fig. 3, left). The six voters' per-token error vectors have mean pairwise correlation 0.74; the four AraBERTv02 seeds are nearly interchangeable. High error correlation is exactly why adding more same-data voters does not help—they err together—and motivates the open-track search for *decorrelated* voters.

(iv) Calibration (Fig. 3, right). The ensemble is well calibrated (expected calibration error 0.005 on PA, 0.020 on NoPnx-NP), which is the precondition for the length prior to help: a prior multiplies probabilities, so it pays off only when those probabilities are trustworthy. This closes the loop with the saturation study, where the prior was inert over weaker single models.

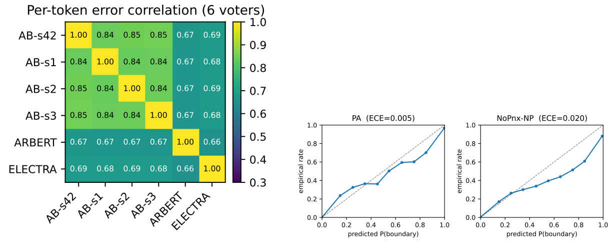

Figure 3: Left: per-token error correlation across the six voters (mean off-diagonal 0.74). Right: reliability diagrams; the ensemble is well calibrated (ECE 0.005/0.020).

7.1 Quantifying the ceiling

The four diagnostics are qualitative; we now put numbers on the ceiling (Table 6). **Perceived vs. true ambiguity.** Because the ensemble is well calibrated, the error of the *optimal* decision on its probabilities—a Bayes floor $\sum_{\text{bins}} n_b \min(r_b, 1-r_b)$—rises with difficulty from 0.85% (PA) to 3.10% (NoPnx-NP), and the share of true boundaries in uncertain bins ($p \in [0.3, 0.7]$) grows from 2.6% to 6.6%. But this is the model's *perceived* ambiguity; is it intrinsic? A model-free check says largely not. When the same $\pm$2-token context recurs in gold, its boundary label is consistent 99.9% (PA) / 99.3% (NoPnx-NP) of the time, so an optimal context-only classifier would err on just 0.05% / 0.21% of repeated-context tokens. The perceived uncertainty thus exceeds the true gold inconsistency by an order of magnitude: most apparent ambiguity is *sparsity*—the model rarely sees a context twice—not irreducible noise. This sharpens the diagnosis: the ceiling is largely reducible with data (consistent with the still-rising scaling curve), and the comma is hard not because it is random but because each comma context is rare. **A power-law asymptote.** Fitting $F_1(n) = \hat{C} - a\,n^{-\alpha}$ to the seed-averaged curve (Fig. 2, left; 2000 parametric-bootstrap fits) gives significantly different exponents: PA α=1.37 $[1.01, 1.74]$ at asymptote $\hat{C}$=94.8 $[94.4, 95.3]$, versus NoPnx-NP α=0.66 $[0.46, 0.83]$ toward $\hat{C}$=86.4 $[85.4, 88.4]$—an asymptote whose entire interval lies above the current 84.5, so in-distribution head-room is statistically confirmed though approached slowly. **An ensemble floor.** A Krogh–Vedelsby decomposition (ensemble error = mean-voter error − diversity) gives mean-voter MSE 0.030 but diversity only 0.0044 (14%): the six same-data voters err together. Adding two external voters raises diversity only to 0.0045, matching the bounded +0.11 F_1 we observe (§10)—more decorrelated data, not more voters, is the lever.

8 Selection Under a Small Development Set

Two configurations—the six-model ensemble with DP decoding and the simpler three-encoder ensemble—differ by ≤ 0.1 F_1 on development; with ∼28 documents per genre this is below the set's resolution. Bootstrap-resampling documents 3000 times (Table 7) gives each configuration's win probability. We adopt the rule: trust the development choice only when its 95% CI for ΔF_1 excludes zero; on a tie, choose the lower-variance configuration (fewer development-fit knobs)—a prior, not a peek at test. This flips two of four blind-test choices (NoPnx-PA, NP) to the three-encoder ensemble; the held-out test validates the methodology without informing it (NP +0.13; NoPnx-PA a wash, lower variance). For small, multi-genre development sets, the simpler of two tied systems is the safer blind-test bet.

Task	Bayes floor	amb. pos. share	$\hat{C}$ (asymp.)	α
PA	0.85%	2.6%	94.8	1.37
NP	1.20%	3.7%	–	–
NoPnx-PA	2.42%	5.9%	–	–
NoPnx-NP	3.10%	6.6%	86.4	0.66

Table 6: Quantified ceiling. Bayes floor: irreducible per-token error of the optimal decision on the calibrated ensemble probabilities. amb. pos. share: fraction of true boundaries with $p \in [0.3, 0.7]$. $\hat{C}, \alpha$: power-law asymptote and exponent of the scaling curve (fit for the two tasks with curves). Difficulty, irreducibility, and slow scaling rise together.

Task	P(3-enc wins)	95% CI ΔF_1	decision
PA	1.2%	$[-0.41, -0.03]$	6-model
NoPnx-PA	28.5%	$[-0.43, +0.26]$	3-enc
NP	8.8%	$[-0.53, +0.07]$	3-enc
NoPnx-NP	2.0%	$[-0.72, -0.02]$	6-model

Table 7: Bootstrap stability on development (3000 resamples). Ties break toward the simpler system.

9 Error Analysis

We dump every residual error on development, split into missed (FN) and spurious (FP) boundaries, and bucket by the token at the boundary and its predecessor; we estimate irreducibility by how often each context is a boundary in training.

Punctuated tasks are comma-bound. On PA and NP, errors concentrate on the comma (the top FN and FP), which is a gold boundary only 5–10% of the time *marginally*. Yet, as §7.1 shows, the decision is near-deterministic given enough context (99.9% label-consistent for repeated $\pm$2-token windows): the comma is hard because each of its contexts is rare, not because it is random. PA's flat scaling curve then reflects that its few recurring contexts are already learned, while its residual is the long tail of unseen ones.

No-punctuation tasks are sparsity-bound. 88–89% of missed boundaries fall on bigrams never seen in training. One bounded systematic exception: the model over-fires on *farāgh* ("blank"), a fill-in-the-blank placeholder producing 80–98 spurious boundaries across four cloze documents of a genre seen twice in training; fixing them perfectly raises dev

Genre	n	PA	N-NP	Δ
general prose	209	95.4	84.7	−10.7
hadīth/isnad	7	91.1	90.8	−0.3
legal/constitution	1	83.1	86.5	+3.4
genealogy	1	100.0	57.5	−42.5
cloze/exercise	4	91.6	70.0	−21.6
all	222	95.15	84.53	

Table 8: Per-genre dev F_1 on identical documents, with vs. without punctuation.

F_1 from 84.53 to only 85.07. The fix is more cloze-genre data, not a development-tuned suppression.

Per-genre punctuation dependence (Table 8). Scoring the *same* documents with versus without punctuation isolates each genre's reliance on punctuation. *Hadīth* chains barely drop (−0.3)—they are lexically self-regular ("*'an X 'an Y*")—while general prose drops −10.7, and genealogy (−42.5) and cloze (−21.6) collapse, encoding structure that punctuation otherwise carries. (Buckets are signature-based heuristics; non-prose counts are small.)

10 Open Track

We synthesise pseudo-documents from external Arabic sentences (a boundary on each sentence-final token; punctuation stripped for no-punctuation conditions), pretrain the boundary classifier on them, and fine-tune on AraSeg training data. Zero-shot transfer is real: a model trained only on external data reaches 0.625 proxy F_1 on development with no in-domain training. But a *single* external voter is flat to negative when added to the closed ensemble—external text supplies structure, not AraSeg's annotation conventions. The operative lever is *diversity*: two external voters from *different* corpora (OPUS subtitles, Ashaar verse) stack to beat the closed ceiling on NoPnx-NP (test 85.08>84.97), exactly the decorrelation the 0.74 voter correlation said was missing.

Data diversity saturates; architecture diversity does not. Pushing the *data* axis further fails on every front: a third same-recipe voter (Wikipedia) plateaus; a classical-register voter (500k Tashkeela sentences, matched to the corpus's hardest genres) and a 1M-sentence scale-up voter are individually solid (83.0/82.6 dev) yet *dilute* the pool (84.55–84.58 $<$ 84.59); and classical data actively hurts NoPnx-PA, showing its immunity to external data is structural, not a register mismatch. What does work is changing the *architecture*: a fine-tuned SaT-12L [6] voter—character-aware, multilingually pretrained for exactly this task—is the *weakest* candidate solo (79.9 dev after fine-tuning, from 68.5 zero-shot) yet the *only* one that improves the pool, the Krogh–Vedelsby prediction in its purest form. Greedy selection then lands on a *four*-voter pool (one AraBERTv02 seed, AraELECTRA, the OPUS voter, and SaT-ft): dev 85.06 (+0.47 over the eight-voter pool, paired bootstrap CI [+0.15, +0.80], so the development evidence passes our §8 adoption rule) and test 85.17>85.08. Without SaT-ft the same search dev-overfits (dev +0.13, test −0.09)—our selection rule rejects it, validating the methodology out of sample. The final open system is thus *smaller* than the closed one: decorrelation, not accumulation, is what the data-limited task buys with external resources.

Pushing the architecture axis. A final round probes how far architecture diversity extends, under the same adopt-only-if-dev-CI-excludes-zero *and* test-confirms rule. A longer-trained SaT (+2.1 solo) and SaT applied to NoPnx-PA both post *positive* signals—the latter significantly on dev (+0.43, CI [+0.13, +0.75])—yet fail test confirmation and are rejected; the re-tuned decode grid likewise offers nothing beyond the frozen parameters. One candidate passes both gates: adding mDeBERTa-v3 [7]—a disentangled-attention encoder, the fourth architecture family we try—to the NoPnx-PA ensemble (dev +0.54, CI [+0.20, +0.87]; test 87.27>87.19). Being an off-the-shelf pretrained encoder fine-tuned only on AraSeg training data, it is closed-track legal, so both NoPnx-PA submissions become four-encoder ensembles. The axis has a sharp boundary condition, however: a *causal* LLM voter (Qwen2.5-3B, LoRA token classifier) collapses to 56 F_1 solo and only harms every pool—boundary detection hinges on the token *after* the candidate position, and no amount of decoder-side pretraining substitutes for right-context. Across $\sim$26 post-freeze attempts, the only two survivors are *bidirectional* architecture changes—consistent with every diagnostic in §7.

11 Related Work

Sentence segmentation has moved from rule and statistical models to neural sequence labelling; multilingual SaT/`wtpsplit` [6] achieves punctuation-robust performance via character-level corruption pretraining. Our setting is low-resource and Arabic-specific, where the organisers find lightweight encoders competitive with large language models in the hardest conditions. We build on Arabic encoders [1, 2, 3, 4] and semi-Markov decoding, and contribute a controlled study of where their gains saturate, with scaling, coverage, correlation, and calibration evidence.

12 Conclusion

A strong ensemble-plus-decoding system led all four development-phase closed-track leaderboards, but our main result is a quantified demonstration that the closed-track ceiling is set by data, not architecture. On the hardest task the scaling curve is still rising toward an 86.4 F_1 asymptote (α=0.66) while only 5% of its boundary contexts are seen in training; the calibrated Bayes floor places 3.10% of tokens beyond reach; and the six voters err at correlation 0.74, so the pool sits at its diversity floor. The easiest task, by contrast, is at its data asymptote with a 0.85% irreducible floor. We contribute a bootstrap procedure for configuration selection under a small development set, this four-way quantified diagnostic of the ceiling, and an open-track result in which external pretraining helps only by adding decorrelated voters, and only on the data-limited task—exactly where the diagnostics predict head-room remains.

Limitations

Conclusions concern one corpus and language and may not transfer to larger Arabic segmentation datasets. The open-test split, though held out for selection, is public, so our verification is weaker than the blind test. Per-genre buckets are heuristic with small non-prose counts. External-data experiments cover five corpora (subtitles, verse, Wikipedia, classical Tashkeela, and a 1M-sentence mix) under one boundary-recovery recipe; other pretraining objectives might transfer differently. The scaling curve uses a single encoder over three seeds; the power-law extrapolation assumes the fitted functional form holds beyond 174 documents, which only new data could verify.

Appendix: Reproducibility

Every run is logged with its configuration and dev/test F_1. Training: `train_encoder.py` (windows, `[PAR]` token, weighted loss, optional label smoothing / FGM). Inference/decoding: `predict.py`, `cache_probs.py`, `dp_decode.py`, `dp_adaptive.py`, `sweep_threshold.py`, `ensemble_sweep.py`. Analysis: `bootstrap_stability.py`, `residual_errors.py`, `genre_buckets.py`, `significance.py`, `make_figures.py`, `scaling_eval.py`, `scaling_bands.py`, `analysis_astar.py`, `analysis_astar2.py`. Open track and post-freeze rounds: `build_pretrain.py`, `open_pipeline.sh`, `build_open2_data.py`, `train_sat_ft.py`, `train_qwen_lora.py`, `open_eval2.py`, `round3_eval.py`, `round4_eval.py`, `open_sweep4.py`. A single label-free entry point, `predict_blind.py`, reproduces every submission from the frozen per-task configuration. Environments: `torch` 2.5.1 / `transformers` 4.57 for `safetensors` encoders, and `torch` 2.12 for `.bin` checkpoints.

References

[1] W. Antoun, F. Baly, H. Hajj. AraBERT: Transformer-based Model for Arabic Language Understanding. LREC OSACT, 2020.

[2] W. Antoun, F. Baly, H. Hajj. AraELECTRA: Pre-Training Text Discriminators for Arabic Language Understanding. WANLP, 2021.

[3] M. Abdul-Mageed, A. Elmadany, E. M. B. Nagoudi. ARBERT & MARBERT: Deep Bidirectional Transformers for Arabic. ACL, 2021.

[4] G. Inoue, B. Alhafni, N. Baimukan, H. Bouamor, N. Habash. The Interplay of Variant, Size, and Task Type in Arabic Pre-trained Language Models. WANLP, 2021.

[5] A. Conneau et al. Unsupervised Cross-lingual Representation Learning at Scale. ACL, 2020.

[6] M. Frohmann, I. Sterner, I. Vulić, B. Minixhofer, M. Schedl. Segment Any Text: A Universal Approach for Robust, Efficient and Adaptable Sentence Segmentation. EMNLP, 2024.

[7] P. He, J. Gao, W. Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-Style Pre-Training with Gradient-Disentangled Embedding Sharing. ICLR, 2023.

[8] T. Miyato, A. M. Dai, I. Goodfellow. Adversarial Training Methods for Semi-Supervised Text Classification. ICLR, 2017.

[9] AraSeg Shared Task Organisers. The AraSeg Corpus for Arabic Sentence Segmentation. arXiv:2606.08025, 2026.